Sasikanth Vadde

vaddesasikanth@gmail.com | linkedin.com/in/sasikanth-vadde | github.com/Sasi-vadde | sasi-vadde.github.io/portfolio | Stuttgart, Germany

PROFESSIONAL SUMMARY

Chemical & Energy Engineer (M.Sc.) specializing in electrochemical and multiphysics modelling of **alkaline water electrolysis (AWE)**. Two years at DLR (German Aerospace Center), progressing from intern to Research Associate and extending a single-cell **Dymola/Modelica** model to a high-pressure stack digital twin. Models validated against test-bench measurements; CFD in **OpenFOAM**; **Python** for data analysis and model prototyping.

KEY SKILLS

Alkaline Water Electrolysis (AWE) | Dymola/Modelica Digital Twins | Electrochemical & Two-Phase Modelling | Parameter Identification | CFD & Conjugate Heat Transfer (OpenFOAM) | Multi-Loop PID Process Control | Measurement-Data & Root-Cause Analysis | Python & Data Analysis

WORK EXPERIENCE

DLR (German Aerospace Center), Institute of Engineering Thermodynamics

10/2023 - 12/2025

Intern, Master's Thesis Student, Working Student, Research Associate

Stuttgart

- Value added: progressed from intern to Research Associate, extending a single-cell AWE model to a **high-pressure stack digital twin** used for system-level KPI assessment.
- Modelling: to simulate transient cell behaviour, built a 1D single-cell AWE model in **Dymola/Modelica** by coupling published electrochemical, two-phase and thermal correlations, then extended it to a **high-pressure stack** and a two-pressure-vessel concept.
- Modelling: to describe gas solubility under pressure, added a **fugacity correction** for non-ideal gas behaviour and reworked the electrolyte-property model (missing functions added, correlations adapted).
- Validation: **validated the transient models against test-bench measurements**; traced a model-vs-measurement mismatch to insufficiently calibrated pump flow-rate measurements and corrected two modelling defects (heat-release assumption, electrolyte density correlation).
- Integration: identified electrochemical kinetic and gas-crossover parameters and built a **hybrid gas-crossover model** (empirical correlations plus data-driven corrections) for DLR's in-house system-level simulation platform; prototyped each new equation in **Python** before integrating it.
- Thermal: to calibrate the Nusselt-number correlations of the thermal stack model, ran conjugate-heat-transfer CFD in **OpenFOAM** (chtMultiRegionFoam) of endplate cooling-channel flow.
- Control: to keep gas purity within safety limits under fluctuating renewable load, developed **multi-loop PID control** strategies; generated ARMA-based wind and PV load profiles in Python to drive the transient simulations.
- Test rigs: supported two AWE test rigs: contributed to P&ID diagrams, helped set up equipment, analysed experiments and helped with test scripts.
- Tools: Dymola/Modelica, OpenFOAM, Python

Otto von Guericke University Magdeburg

06/2023 - 08/2023

Working Student (HiWi), Chair of Mechanical Process Engineering (Prof. Berend van Wachem)

Magdeburg

- Image analysis: to calibrate a drying-process model, built a **Python/OpenCV** bubble-tracking pipeline for fluidized-bed images.
- Experiments: ran fluidized-bed experiments with varying inlet pressure and sinusoidal airflow on the test rig.
- Tools: Python, OpenCV

GMR Hyderabad International Airport (GMR Group)

07/2019 - 03/2021

Graduate Engineer Trainee - Enforcement Officer, Airside Operations

Hyderabad, India

- Training: started with cross-functional training, including a visit to observe the construction of the Bajoli Holi hydropower plant.
- Airside operations: as Enforcement Officer, received multiple **'Golden Card' awards** for handling emergencies.

Hindustan Aeronautics Limited (HAL)

11/2018 - 04/2019

Full-semester Internship

Koraput, India

- Quality: to assess process stability, analysed manufacturing, inspection and fatigue-testing sequences for AL-31FP turbine blades using **statistical process control** (X-bar and S charts).

PROJECTS

Electrochemical-Thermal Modelling of a Li-ion Cell

Physics-based **P2D (Doyle-Fuller-Newman)** model of a commercial 21700 Li-ion cell with porous electrodes (Battery Design Module), parameterised from published measurement data and literature values, with evaluation in Python. Next: electrochemical-thermal coupling and a cross-check with PyBaMM.

COMSOL Multiphysics, Python, since 10/2026

CFD Simulations

Wind flow around the Arc de Triomphe (aerodynamic loading, stress hotspots). Natural convection around rectangular heating elements (buoyantPimpleFoam, mesh-sensitivity and boundary-condition study).

STAR-CCM+, OpenFOAM, 2022-2023

Multivariable Process Control

PID tuning of the Newell-Lee evaporator (Ziegler-Nichols, Cohen-Coon, IMC); step-response and integral-error evaluation.

MATLAB/Simulink

EDUCATION

M.Sc. Chemical and Energy Engineering Otto von Guericke University Magdeburg

04/2021 - 10/2025

Magdeburg

Grade 1.9. Thesis: Transient Modelling of an Alkaline Water Electrolyzer for Validating a Two-Phase Electrolyte Model (MPI Magdeburg + DLR Stuttgart). Focus: CFD, Process Control & Systems Engineering, Multiphase Flows.

B.Tech Mechanical Engineering GMR Institute of Technology

07/2015 - 04/2019

Rajam, India

Focus: Thermodynamics & Heat Transfer, Fluid Mechanics, FEM, Power Plant Engineering & Energy Systems.

TECHNICAL SKILLS & LANGUAGES

Simulation: Dymola/Modelica, MATLAB/Simulink, OpenFOAM, STAR-CCM+, ANSYS, COMSOL Multiphysics, Aspen Plus, DWSIM

Modelling: electrochemical kinetics, two-phase/multiphase flow, heat & mass transfer, coupled multiphysics

Process & control: PID & multi-loop control, P&ID/PFD, PLC (IEC 61131-3), measurement-data & root-cause analysis

Programming & data: Python (PyTorch, scikit-learn, NumPy, Pandas, OpenCV), C++, MATLAB, Git, openLCA, AutoCAD

Languages: German (B1), English (C1/C2), Telugu (native), Hindi (native-level)